Data Requirements

Automated Vehicle Damage Classification for Motor Insurance Claims | Bul Ins

Project phase: After Domain Understanding · Before dataset selection

1. Project Goal and Analytic Approach

The project aims to support the initial triage of motor insurance claims at Bul Ins in Bulgaria by identifying visible vehicle damage from photographs. A claims handler or inspector remains responsible for assessment and claim decisions. The initial analytic approach is supervised image classification, with damage type as the primary target. Recognising the affected vehicle part is a potential complementary task, subject to data availability and feasibility. The model should return a predicted category and confidence score; it must not determine repair costs or total-loss status from an image alone.

2. Data Format and Type

Required inputs are digital photographs of damaged vehicles (unstructured image data) and corresponding structured labels/annotations. Images should be readable in standard formats such as JPEG or PNG. A structured annotation file (for example CSV or JSON) should link each image to its verified label(s) and, where available, a bounding box or segmentation mask. A unique image identifier is needed for traceability.

3. Data Representation

Element	Requirement / reason
Damage photograph	Clearly shows the relevant damage; varying angles, lighting, backgrounds and vehicle types are desirable for generalisation.
Damage-type label	Mandatory supervised-learning target. Candidate categories include dent, scratch, crack, broken part, paint damage, missing part and corrosion; final taxonomy depends on domain relevance and selected dataset.
Vehicle-part label	Desirable additional target (e.g., bumper, door, windshield, headlight) to identify which component is affected. Not guaranteed in the initial dataset.
Annotation link	Each label must map to the correct image or annotated damage region. Multiple damages per image must be handled

	explicitly.
Image metadata	Image dimensions, file format, source, annotation provenance and any permitted capture information to support quality checks.

Business rule: the dataset's labels must be compatible with the intended task. If the source provides multiple damage instances per image, the project must decide whether to crop instances, use multi-label learning or define a justified single-label rule before training.

4. Data Quantity

No dataset has been selected yet, so the available image count and class distribution are unknown. Dataset selection will favour enough labelled examples in each category for meaningful training and evaluation, not just a high total image count. The minimum acceptable size will be assessed after exploratory analysis and an initial baseline; transfer learning and data augmentation may help if data is limited. Training, validation and test sets must be separated, with a proposed starting split of 70% / 15% / 15%, adjusted for class balance and dataset size. Images from the same original vehicle/claim or near-duplicate sequence must not leak across splits.

5. Data Quality Requirements

Accuracy: Damage labels must correctly describe visible damage; uncertain annotations should be reviewed.

Completeness: Required images and labels must be present, with missing or unreadable files identified.

Consistency: Category names, annotation formats, and preprocessing rules must be standardised.

Uniqueness: Detect exact and near-duplicate images to prevent misleading evaluation.

Validity: Verify readable file formats, image dimensions, permitted labels and valid annotation coordinates.

Timeliness: Check whether image examples and vehicle types are relevant to the intended current claims workflow.

During Data Understanding, examine class imbalance, image resolution, blur, occlusion, lighting, angles, vehicle diversity, incorrect labels and potential distribution differences between public images and Bul Ins claim photos.

6. Legal and Ethical Constraints

Before choosing or using a dataset, verify its origin, licence, redistribution rights and permitted academic use. Check for visible faces, licence plates, documents or other

personal data and minimise or anonymise these where necessary. Any future Bul Ins claims data requires a lawful basis, appropriate access controls, retention rules and GDPR assessment. The system should support human reviewers, communicate uncertainty, and be evaluated for unequal performance across damage types and vehicle conditions. Applicable EU AI Act obligations should be assessed for the actual deployment use case rather than assumed from the model alone.

7. Dataset Selection Criteria (Next Step)

Potential datasets will be compared against: (1) relevance to real insurance damage photos; (2) clear and reliable damage-type labels; (3) sufficient examples per class; (4) image and annotation quality; (5) legal usability; (6) availability of part labels or region annotations; and (7) compatibility with practical model training. The final dataset name, source, class list, counts, split and limitations will be documented only after selection and exploratory data analysis.

8. Future Extension: Repair Cost Estimation

Repair-cost estimation is not part of the initial image-classification deliverable. A later model would require additional structured data beyond photographs: vehicle brand, model, production year and trim; identified damaged part; damage severity; replacement-part prices (including OEM/aftermarket distinctions); labour rates and repair hours; paint/material costs; regional price variation; historical claims with actual paid repair amounts; vehicle market value and insured value. Dates and currencies would need standardisation, and costs should be matched to the correct vehicle, component and damage case.

These data could support a repair-cost estimate and a comparison with vehicle value for human review, but would not independently establish an insurer’s total-loss decision. Historical pricing and claim data would require separate sourcing, permissions and validation.

9. Planned Data Understanding Activities

After selecting a dataset: document its provenance and licence; count samples per class; inspect image and label quality; check duplicates and sensitive content; establish preprocessing and annotation rules; create leakage-safe splits; and record limitations and improvements for the next AMD iteration.

Basis: AI Project Proposal (September 2026), Domain Understanding – Bul Ins, and Fontys ICT Data Requirements workshop. Dataset-specific figures are intentionally omitted because selection has not yet occurred.